Do Vision Transformers Need Three Projections? Scaling Projection-Sharing Attention to ImageNet

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Draft — August 26, 2026

Abstract

We study whether the projection-sharing attention variants shown to preserve language-model quality while halving the KV cache [Kayyam et al., 2026] generalize to large-scale vision transformers. Prior work in this line — including our own systematic study of query/key/value projection sharing on synthetic, vision (MNIST/CIFAR/TinyImageNet), and language-modeling tasks, and a concurrent line of work reframing the key-value-shared variant as "Keyless Attention" with a value-space routing interpretation [Gao and Xu, 2026] — has validated projection sharing at small-to-medium model and dataset scale. It remains open whether these gains persist at ImageNet scale, and whether the overfitting-reduction effect attributed to gradient entanglement between routing and retrieval parameters [Gao and Xu, 2026] is a language-specific artifact or a general property of shared-projection attention. We train ViT-S/16 from scratch on full ImageNet-1k under a compressed DeiT-style recipe, once each, for the QKV baseline and three projection-sharing variants (Q-K=V, Q=K=V$^+$, QVV(3)). Q-K=V and QVV(3) close most of the gap to the QKV baseline (71.7% and 72.5% top-1 vs. 74.4%), while the fully symmetric Q=K=V$^+$ variant lags substantially (66.8%) — a larger degradation, in relative terms, than the 3.1% perplexity cost reported for the language-modeling setting, and one that does not support the hypothesis that symmetric attention maps are less costly in vision than in causal language modeling. We find no measurable overfitting-gap difference between variants at this schedule: the heavily regularized 100-epoch DeiT recipe (RandAugment, Mixup, CutMix, stochastic depth, label smoothing) leaves final- and best-epoch validation loss within noise of each other for every condition, including the QKV baseline, so the gradient-entanglement regularization effect reported for language models is neither confirmed nor refuted here. A follow-up run (Section 5.2) closes a gap flagged in the original design — the Q=K-V$^+$ condition, previously skipped to fit four conditions across four GPUs — and adds an unplanned ablation removing the 2D positional correction from Q=K=V$^+$. Q=K-V$^+$ reaches 70.4% top-1, trailing its asymmetric analog Q-K=V by only 1.3 pts (vs. Q=K=V$^+$'s 4.9-pt gap), suggesting the earlier symmetric-attention penalty is driven more by collapsing V into the shared projection than by attention-map symmetry itself; dropping the positional correction entirely raises Q=K=V from 66.8% to 70.7%, though this run also doubled the effective batch size, so the two effects are confounded and not yet separable. A ViT-B/16 confirmatory run, an ImageNet-100 pilot phase, and a downstream detection-transfer evaluation were part of the original design but were descoped for this iteration; we report what a single-scale, single-seed ViT-S/16 study on ImageNet-1k shows and leave the rest as future work.

1 Introduction

The query-key-value (QKV) formulation of self-attention [Vaswani et al., 2017] is near-universal in both language and vision transformers, yet the individual necessity of each of the three projections is not well understood. We previously showed that in language models, merging the key and value projections ($K = V$, which we term **Q-K=V**), merging the query and key projections with a corrective 2D positional term (**Q=K-V^{+}**), and even collapsing all three into a single projection can match or nearly match standard QKV attention, with the $K=V$ variant achieving a 50% KV-cache reduction at only 3.1% perplexity degradation on 300M–1.2B parameter language models trained on 10B tokens [Kayyam et al., 2026]. That study also included a preliminary vision pass on MNIST, CIFAR, and TinyImageNet.

A concurrent paper, *Keyless Attention* [Gao and Xu, 2026], independently arrives at the same base architecture as our Q-K=V variant — algebraically, $\mathrm{softmax}(QK^{\top})K \equiv \mathrm{softmax}(QV^{\top})V$ when $K = V$ — but frames it as eliminating the key projection and routing directly through value space, rather than as a special case of projection sharing. That paper extends the idea along a new axis, *depth-m attention factorization*, in which the query and/or value projection is itself a composition of m learned matrices, and shows that the depth-3 instantiation (**QVV(3)**) matches the parameter count of standard QKV attention while still requiring only a value cache at inference time. It also reports a consistent finding across five language models and four architectures (GPT-2, Pythia, Qwen2, Llama 3.2): projection-shared attention degrades *more slowly* after the best-validation epoch than standard attention, and attributes this to a gradient-entanglement effect between the routing and retrieval parameters that acts as an implicit regularizer. All five of their experiments are causal language models; QVV(2) and QVV(3) are never evaluated on a vision transformer, so the equivalence to our Q-K=V variant is algebraic rather than empirically confirmed on non-causal, 2D-grid attention — the gap this paper's ImageNet-1k experiments address.

Two questions follow naturally from combining these two lines of work, and motivate this paper:

1. **Does projection sharing hold at ImageNet scale?** All vision evidence to date for this family of methods is at MNIST/CIFAR/TinyImageNet scale, with small models. Vision transformers at the scale used in production (ViT-B/16, ViT-L/16, ImageNet-1k) have a very different data/parameter regime, and it is not obvious that a 5.6%-perplexity-scale finding on WikiText-103 will transfer.

2. **Is the overfitting-reduction effect language-specific?** The gradient-entanglement argument in Gao and Xu [2026] is architecture-agnostic in principle, but has only been tested on autoregressive causal language modeling. Vision transformers use bidirectional (non-causal) attention over a 2D patch grid, which changes both the routing geometry and the overfitting dynamics (ImageNet-1k, at 1.28M images, is also a substantially larger and more diverse training set relative to model capacity than WikiText-103 at equivalent scale).

We additionally note that vision transformers offer a structural advantage absent in language: patches sit on an explicit 2D grid, so the 2D positional-encoding correction we introduced for the symmetric Q=K-V and Q=K=V variants in Kayyam et al. [2026] is a *native* fit rather than an artificial addition, unlike in the 1D sequential setting of language modeling.

This paper originated as a design document laying out architectures, training recipes, compute budget, hypotheses, and evaluation protocol for a multi-scale study (Section 4 still describes that original plan). For this iteration, the scope was deliberately narrowed to a single architecture (ViT-S/16), a single training run per condition, directly on full ImageNet-1k with no ImageNet-100 pilot phase, and no detection-transfer experiment — prioritizing a complete, correctly-run single-scale result over a partially-completed multi-scale one. Section 5 reports the resulting numbers; Section 7 discusses what the descoping leaves untested.

1.1 Contributions

- Evaluation of three projection-sharing attention variants (Q-K=V, Q=K=V^{+}, and QVV(3)) against a standard QKV baseline on ViT-S/16 trained from scratch on full ImageNet-1k, extending prior evidence for this family of methods beyond MNIST/CIFAR/TinyImageNet scale.
- A direct test of whether the overfitting-reduction / gradient-entanglement effect reported for language models [Gao and Xu, 2026] reproduces under non-causal, 2D-grid attention — which, at this training schedule, it does not measurably do, for any variant including the baseline (Section 6).
- A follow-up run adding the previously-skipped Q=K-V^{+} condition and an unplanned ablation of the 2D positional correction on Q=K=V (Section 5.2), refining the H3 verdict from Section 6.
- *Deferred to future work:* a ViT-B/16 (and ViT-L/16) confirmatory run at larger scale, an ImageNet-100 pilot phase, and a downstream detection-transfer evaluation (FCOS/GFL-style), all part of the original design (Section 4) but out of scope here.

2 Related Work

Projection sharing in attention. Kayyam et al. [2026] systematically evaluate three projection-sharing constraints — Q=K-V, Q-K=V, and Q=K=V — and show that the last two, which produce symmetric attention maps, can be de-symmetrized via a 2D sinusoidal positional correction. The Q-K=V variant is highlighted as the most favorable trade-off, achieving 50% KV-cache reduction with a 3.1% perplexity cost. Gao and Xu [2026] re-derive the same base mechanism from a "value-space routing" motivation and generalize it via depth-m attention factorization, introducing the QVV(3) variant that matches standard attention's parameter count and, in their experiments, matches or exceeds it on perplexity and zero-shot benchmarks across five models while halving the KV cache. Borji [2023] is an earlier, independent precursor examining a KV-only (query-key merged) transformer formulation.

Shared projections in vision transformers. Lee et al. [2021] share nonlinear embedding layers across Q, K, V in tiny ImageNet-scale ViTs (3.1M parameters) with a code-parameterized sharing scheme, reporting 71.4% top-1 accuracy at small scale. MASA [MASA, 2025] decomposes attention projection matrices into a shared dictionary of matrix atoms, cutting attention parameters by 66.7% with on-par performance, and includes a ViT classification experiment. Neither targets KV-cache reduction or ViT-B/L scale training on full ImageNet-1k; both motivate parameter reduction rather than routing/retrieval unification, and neither reports an overfitting/regularization analysis. We view this as the primary novelty gap this study addresses on the vision side.

KV-cache reduction more broadly. Multi-Query Attention [Shazeer, 2019] and Grouped-Query Attention [Ainslie et al., 2023] reduce the KV cache by sharing keys/values across attention heads, and Multi-Head Latent Attention [DeepSeek-AI, 2024] compresses keys and values into a shared low-rank latent. These methods are orthogonal to projection sharing, which instead removes redundancy between the key and value *projections themselves*; Kayyam et al. [2026] and Gao and Xu [2026] both note the two families compose.

ViT training recipes. We adopt training conventions from Dosovitskiy et al. [2021] and Touvron et al. [2021]; DeiT's data-efficient training recipe (RandAugment, Mixup, CutMix, stochastic depth, label smoothing) is essential for training ViT-B from scratch on ImageNet-1k without ImageNet-21k pretraining, and we use it as our default recipe with a compressed epoch budget (Section 4).

3 Method

3.1 Recap: projection-sharing variants

Standard multi-head attention computes, per head,

$$\mathrm{Attn}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \qquad Q = XW^Q,\ K = XW^K,\ V = XW^V. \tag{1}$$

We evaluate the following variants, applied to a standard ViT patch-embedding backbone (Section 4):

Q-K=V. Set $V = K$, i.e. drop W^V and read the output from the key projection: $A = \mathrm{softmax}(\alpha QK^\top)K$. Equivalent to the QVV(2) base case of Gao and Xu [2026]. Halves the cache (single projection cached in the autoregressive setting; for ViT, halves activation memory for the shared K/V tensor during training and any cached-inference use case, e.g. streaming or tiled inference).

Q=K-V$^+$. Set $Q = K$, producing a symmetric attention map $\mathrm{softmax}(\alpha KK^\top)V$; corrected for asymmetry with a 2D positional term (Section 3.2).

Q=K=V$^+$. Collapse all three projections to one, with the same 2D positional correction.

QVV(3). Depth-3 factorization from Gao and Xu [2026]: $Q = XW^{Q_1}W^{Q_2}$, $V = XW^V$, output $\mathrm{softmax}(QV^\top/\sqrt{d_k})V$. Matches standard attention's parameter count; at inference, $W^{Q_1}W^{Q_2}$ is fused into a single matrix, so it incurs no extra cost over Q-K=V while retaining full QKV parameter capacity.

All variants preserve the standard ViT patch embedding, learned or sinusoidal position embedding for the base sequence order, MLP block, LayerNorm placement, and classification head; the attention block is the sole controlled variable, matching the ablation design of both source papers.

3.2 2D positional correction on a native 2D grid

Kayyam et al. [2026] introduce a 2D sinusoidal positional correction to de-symmetrize the Q=K-V and Q=K=V attention maps in language modeling, where the 2D structure is an artificial construction imposed on a 1D token sequence. In ViT, patches are natively arranged on a 2D grid ($H/p \times W/p$ for patch size p), so the same correction can be applied using the model's actual patch coordinates rather than a synthetic 2D reindexing of a 1D sequence:

$$\tilde{S}_{ij} = S_{ij} + \mathrm{Conv}_{1\times1}\big(P(\mathrm{row}_i, \mathrm{col}_i, \mathrm{row}_j, \mathrm{col}_j)\big), \tag{2}$$

where P is a fixed 2D relative-position sinusoidal embedding indexed by true patch row/column, and S is the symmetric raw attention score matrix. We view this as a stronger test of the correction's design intent than the language setting, and a natural point of comparison to existing 2D relative position bias schemes in windowed vision transformers (e.g. Swin [Liu et al., 2021]), which we will cite as a baseline design pattern rather than a competing method.

3.3 Hypotheses

H1 (Accuracy parity). Q-K=V and QVV(3) match standard ViT-B/16 top-1 accuracy on ImageNet-1k within noise ($\lesssim$0.5 pt), consistent with the language-modeling degradation being small (3.1%) and vision transformers being comparatively more over-parameterized relative to ImageNet-1k than LLMs are relative to their pretraining corpora.

H2 (Overfitting reduction transfers). Shared-projection variants show a smaller train/val generalization gap than standard QKV at matched epoch count, replicating the gradient-entanglement regularization effect from Gao and Xu [2026] outside the causal LM setting.

H3 (Symmetric variants underperform more in vision than language). Because 2D spatial relationships in images are less inherently directional than causal token order in language, we expect the gap between corrected symmetric variants (Q=K-V$^+$, Q=K=V$^+$) and the asymmetric variants (Q-K=V, QVV(3)) to be *smaller* in vision than was reported in Kayyam et al. [2026] for language — i.e. symmetry is less costly for spatial patch attention than for causal sequence attention. This is falsifiable and could go the other way if fine-grained spatial ordering (e.g. for texture or object boundary localization) turns out to matter more than expected.

H4 (Transfer to detection is architecture-sensitive). We expect accuracy parity on classification to *not* guarantee parity on detection transfer, since detection localization loss is more sensitive to fine spatial attention structure; this is the primary motivation for including the downstream transfer experiment (Section 4.4) rather than treating ImageNet top-1 as sufficient evidence.

4 Experimental Setup

4.1 Architectures

- **ViT-S/16** (22M params, 12 layers, hidden 384, 6 heads): primary model for the variant sweep, chosen for fast iteration.
- **ViT-B/16** (86M params, 12 layers, hidden 768, 12 heads): confirmatory run on the winning variant(s) from the sweep, and the scale most directly comparable to production vision-backbone usage.
- **ViT-L/16** (307M params): stretch goal, only if the ViT-S $\rightarrow$ ViT-B trend is informative and compute allows; used to test whether any observed effect (H1–H3) grows or shrinks with scale, mirroring the depth-scaling analysis in Kayyam et al. [2026] (12-layer vs. 36-layer GPT-2).

4.2 Datasets and schedule

Two-phase plan to manage compute cost on 1–2 A100-class GPUs:

1. **Pilot sweep — ImageNet-100** (100-class subset, ∼130K images): all five conditions (QKV baseline, Q-K=V, Q=K-V$^+$, Q=K=V$^+$, QVV(3)) on ViT-S/16, 100 epochs, DeiT recipe. Used to pick the variant(s) that proceed to the confirmatory run and to get an early read on H2/H3.

2. **Confirmatory run — ImageNet-1k** (1.28M images): QKV baseline vs. the 1–2 best pilot variants, on ViT-B/16 (and ViT-L/16 if feasible), 100–150 epoch compressed DeiT recipe (RandAugment, Mixup α=0.8, CutMix α=1.0, stochastic depth, label smoothing 0.1), rather than the full 300-epoch recipe, following the same "controlled comparison over SOTA-chasing" framing used for the WikiText-103 subset in Kayyam et al. [2026].

Table 1: Estimated wall-clock training budget (bf16, DeiT-style recipe, data pipeline assumed non-bottlenecked via FFCV/DALI). Two-GPU figures assume ∼1.8× scaling from single-GPU due to gradient-sync overhead.

Dataset	Model	Epochs	1× A100	2× A100
ImageNet-100	ViT-S/16	100	5–7 hrs	∼3 hrs
ImageNet-100	ViT-B/16	100	15–20 hrs	8–10 hrs
ImageNet-1k	ViT-S/16	100–150	12–18 hrs	6–9 hrs
ImageNet-1k	ViT-B/16	100–150	2–3 days	1–1.5 days
ImageNet-1k	ViT-B/16	300 (full DeiT)	6–8 days	3–4.5 days

With 5 conditions in the ImageNet-100 pilot and 2–3 in the ImageNet-1k confirmatory phase, total estimated compute is on the order of ∼4–5 GPU-days on a single A100, or ∼2.5–3 GPU-days on two, dominated by the ImageNet-1k ViT-B/16 runs.

What was actually run. The plan above was descoped before execution: we ran only the ImageNet-1k phase, only on ViT-S/16, and only one seed per condition, skipping the ImageNet-100 pilot and the ViT-B/16/ViT-L/16 confirmatory scale-up. Of the five conditions in the design, four were run — QKV baseline, Q-K=V, Q=K=V$^+$, and QVV(3) — omitting Q=K-V$^+$ (query-key merged with positional correction) to keep the sweep to what fits on four GPUs at once. Each condition trained for 100 epochs on the compressed DeiT recipe of Section 4 (RandAugment, Mixup α=0.8, CutMix α=1.0, stochastic depth, label smoothing 0.1), batch size 128 (the largest that fit in 11GB), on hardware substantially below the A100-class budget in Table 1: four RTX 2080 Ti GPUs, one condition per GPU, run to completion in parallel. Measured throughput was ∼575 img/s/GPU, ∼3320s/epoch, i.e. ∼92 GPU-hours per condition; running all four conditions concurrently, one per GPU, kept total wall-clock to the same ∼92 hours (∼3.8 days) rather than requiring 4× that.

Follow-up run. A second batch, reported in Section 5.2, trained the two remaining conditions: Q=K-V$^+$ (omitted above) and an unplanned ablation, Q=K=V with the 2D positional correction removed. With only two conditions to run and all four GPUs free, each was trained 2-GPU DDP (batch size 128 per GPU, effective global batch 256, base LR linearly scaled to match) rather than the single-GPU, batch-128 setup used for the other four conditions — roughly 2× the effective batch size and LR of Table 2, with epoch time falling to ∼1630–1645s accordingly. This is a genuine setting difference, not just a resource-allocation detail, so Section 5.2's numbers are not a controlled addition to Table 2; we flag comparisons against it explicitly where the batch-size difference matters.

4.3 Evaluation metrics

- Top-1 / top-5 accuracy at best-validation epoch (mirrors Table 1 of Kayyam et al. [2026] and Table 1 of Gao and Xu [2026]).
- Overfitting gap Δ = (final-epoch val. loss) − (best-epoch val. loss), directly testing H2.
- Activation/cache memory reduction for the shared-projection variants, reported both in absolute terms and as a fraction of standard-attention memory (target: 50% for Q-K=V and QVV(3), as in prior work).
- Throughput (images/sec, batch size fixed) at inference, matching the decode-throughput benchmarking methodology of Gao and Xu [2026] adapted to non-autoregressive ViT inference.

4.4 Downstream transfer: detection

To test H4, we fine-tune the ImageNet-1k-pretrained ViT-B/16 backbones (QKV baseline and best-performing shared variant) into a detection pipeline combining FCOS-style dense prediction, GFL loss, and SimOTA label assignment, consistent with our existing detection scaffold. We report COCO-style AP, AP_{50}, AP_{75}, and size-stratified AP (small/medium/large), since we expect any localization sensitivity from projection sharing to appear disproportionately on small-object AP, where fine spatial routing matters most.

5 Results

5.1 ImageNet-1k, ViT-S/16

Table 2 reports top-1/top-5 accuracy at the best validation epoch, the overfitting gap Δ (final-epoch minus best-epoch validation loss), inference-time KV/activation cache relative to the QKV baseline, parameter count, and per-GPU training batch size, for six conditions trained to 100 epochs on full ImageNet-1k. Best-epoch numbers (not final-epoch) are used for Top-1/Top-5 since a small amount of post-peak drift is visible for every condition; final-epoch numbers, used only for Δ, differ from the best-epoch numbers by at most 0.03 pts top-1 for any condition. The first four rows are the original controlled sweep, one condition per GPU at batch 128; the last two (below the rule) are a follow-up batch trained together at effective batch 256 (2-GPU DDP, linearly-scaled LR) rather than batch 128, discussed separately in Section 5.2 because of that setting difference.

Relative to the QKV baseline, Q-K=V and QVV(3) close most but not all of the gap (-2.71 and -1.83 pts top-1), while Q=K=V$^+$ lags well behind (-7.62 pts). Every ΔOverfit value is within 0.0014 nats of zero — an order of magnitude smaller than would be needed to distinguish variants, and consistent with none of the six models overfitting in any meaningful sense at this schedule (see Section 6). Cache and parameter figures match the design targets in Section 4: QVV(3) matches QKV's parameter count while retaining Q-K=V's 50% inference cache, and Q=K=V$^+$ and Q=K=V (no $^+$) tie for the fewest parameters of any condition tested by dropping two of three projections.

5.2 Follow-up: Q=K-V$^+$ and a positional-correction ablation

The last two rows of Table 2 are the two conditions trained in the follow-up batch described in Section 4: Q=K-V$^+$, the condition omitted from the original four-way sweep, and Q=K=V (no $^+$), an ablation that removes the 2D positional correction from Q=K=V$^+$ while leaving every other

Table 2: ImageNet-1k, ViT-S/16, 100 epochs, one run per condition. Top-1/ Top-5 at best validation epoch; Δ Overfit is final-epoch minus best-epoch validation loss (negative values indicate validation loss was still falling at epoch 100). The last two rows trained at a different effective batch size (Section 5.2) and are not directly controlled against the first four.

Variant	Top-1	Top-5	Δ Overfit	Cache	Params	Batch
QKV (baseline)	74.38%	91.93%	+0.0010	100%	22.05M	128
Q-K=V	71.67%	90.32%	−0.0004	50%	20.28M	128
Q=K=V$^+$	66.76%	87.14%	−0.0014	33%	18.50M	128
QVV(3)	72.55%	90.81%	+0.0009	50%	22.05M	128
Q=K-V$^+$	70.40%	89.47%	−0.0011	50%	20.28M	256
Q=K=V (no $^+$)	70.74%	89.74%	−0.0027	33%	18.50M	256

projection unchanged. Both trained 2-GPU DDP at effective batch 256 (vs. batch 128 for the first four rows), so absolute numbers here are not directly controlled against them.

Closing the H3 gap. Q=K-V$^+$ and Q-K=V have identical parameter counts and cache targets (Table 2), differing only in whether Q is a separate projection (Q-K=V) or tied to K (Q=K-V$^+$, corrected for the resulting symmetric attention map). Their gap is 1.27 pts (71.67% vs. 70.40%), far smaller than Q=K=V$^+$'s 4.91-pt gap to the same Q-K=V reference. Since Q=K=V$^+$ differs from Q=K-V$^+$ by also collapsing V into the shared projection, this points at *that* collapse — not the symmetric attention map itself — as the main source of the penalty reported in Section 6, more consistent with H3's original motivation than the all-symmetric-variants verdict the single Q=K=V$^+$ data point supported. The batch-size caveat above applies: Q=K-V$^+$ and Q-K=V were not trained under matched settings, so this reading should be treated as suggestive pending a same-batch confirmatory run.

Positional correction ablation. Removing the 2D positional correction raises Q=K=V's top-1 from 66.76% (Q=K=V$^+$, Table 2) to 70.74% — the opposite of what the correction was designed to do. This is the largest single swing in Table 2, but it is fully confounded with the batch-256/higher-LR setting (Section 4): we cannot currently attribute it to the correction, the batch size, the LR, or an interaction among them. We report it because of its size, not because we can explain it; isolating the cause needs a same-batch-size run of Q=K=V$^+$ (or, symmetrically, Q=K=V at batch 128), left for future work.

5.3 Post-hoc K=V surgery and distillation recovery

Table 2 asks whether *jointly training* Q-K=V from a random initialization matches QKV. A companion question, developed in a parallel note on post-hoc weight surgery (same repository, `distillation` branch), is different: can an already-trained QKV teacher's independent K/V projections be tied together *after the fact*, with no retraining, and if that collapses (it does, everywhere it has been tested so far — synthetic sequence tasks, MNIST/FMNIST/CIFAR-10/100, and a 300M-parameter FineWeb-Edu language model), how much of the gap does a short distillation fine-tune against the QKV teacher's soft logits recover, relative to the jointly-trained Q-K=V ceiling in Table 2? We repeat that protocol here as a first ImageNet-1k/ViT-S/16 data point, reusing the QKV and Q-K=V checkpoints already trained for Table 2 rather than training anything new:

three surgery modes (KEEP_K: student's shared projection $\leftarrow$ teacher's W^K; KEEP_V: $\leftarrow W^V$; AVG: $\leftarrow \frac{1}{2}(W^K + W^V)$), each evaluated zero-shot immediately after surgery, then fine-tuned for 5 epochs against the QKV teacher's soft logits (cross-entropy on hard labels + KL divergence at T=2, equal weighting), with light augmentation (RandomResizedCrop + flip only — the heavy DeiT recipe of Section 4 is for training 100 epochs from scratch, not a short recovery fine-tune, and Mixup/CutMix's soft labels do not compose with hard-label CE).

Table 3: Post-hoc K=V surgery on the ImageNet-1k QKV teacher (74.38% top-1), zero-shot and after 5 epochs of distillation, against the jointly-trained Q-K=V ceiling (71.67% top-1) from Table 2.

Mode	ZS Top-1	ZS Top-5	Distilled Top-1	Distilled Top-5	Δ vs. ceiling
KEEP_K	0.08%	0.46%	71.01%	89.98%	−0.66 pt
KEEP_V	1.91%	6.37%	71.93%	90.43%	+0.26 pt
AVG	0.66%	2.50%	71.87%	90.42%	+0.20 pt

All three collapse to near-random (1/1000 = 0.1% chance level) under zero-shot surgery, consistent with every prior setting and with the convex-hull argument motivating that note: forcing $K = V$ post-hoc confines each attention block's output to the convex hull of vectors an already-specialized K-projection was never trained to preserve query-independent content in. The zero-shot ordering does not replicate the small-scale/LLM pattern where AVG was uniformly safest: here KEEP_V (1.91% top-1) is safest, with AVG (0.66%) between it and KEEP_K (0.08%) rather than above both.

After 5 epochs of distillation, though, the picture that matters more — recovery relative to the from-scratch ceiling — looks *better* here than at any prior scale, not worse: all three modes land within 0.7 pt of the 71.67% ceiling, and KEEP_V and AVG both slightly *exceed* it (by 0.26 and 0.20 pt respectively), echoing the one exception already on record at small scale (AVG/CIFAR-10 also edged past its from-scratch ceiling). Most notably, KEEP_K — the mode with a *persistent*, only-half-closed gap even after 10 epochs on CIFAR-10/100, and the worst-recovering mode at LLM scale too — reaches 71.01% here, just 0.66 pt behind, with per-epoch gains (+51.1, +7.3, +5.6, +4.6, +1.3 pts) decelerating sharply enough by epoch 5 that we did not judge a 10-epoch extension necessary. Both the zero-shot severity and the post-distillation KEEP_K penalty that motivated treating it as the qualitatively harder recovery case at every prior scale are much smaller at ImageNet-1k/ViT-S/16 scale — an open question for why (more distillation steps per epoch on a 1.28M-image training set than on CIFAR's 50k, a wider/deeper model with more redundant capacity to route around a mismatched basis, or something scale-specific to vision classification) that this single run cannot settle on its own.

5.4 ViT-B/16 confirmatory run

Not run in this iteration; deferred to future work along with the ImageNet-100 pilot phase (Section 4).

5.5 Detection transfer

Not run in this iteration. No FCOS/GFL detection scaffold currently exists in this codebase, and standing one up was judged out of scope alongside the scale-up to ViT-B/L; H4 remains untested. Deferred to future work.

6 Discussion

H1 (accuracy parity) — not supported at this schedule. None of the three shared-projection variants comes within the $\lesssim$0.5 pt target. In relative terms, Q-K=V's 2.71 pt drop (3.6% relative) is close to the 3.1% perplexity cost reported for the analogous language-modeling variant in Kayyam et al. [2026], and QVV(3)'s 1.83 pt drop (2.5% relative) is similarly in that range — but Gao and Xu [2026] report QVV(3) *matching or exceeding* standard attention on language perplexity, which does not reproduce here for ImageNet top-1. Q=K=V$^+$'s 7.62 pt drop (10.2% relative) clearly exceeds the LM-scale gap. We cannot rule out that a full 300-epoch DeiT schedule (rather than the 100-epoch compressed one used here) would narrow these gaps, since shared-projection variants may simply need more optimization steps to reach the same optimum.

H2 (overfitting reduction) — untestable at this schedule, not confirmed or refuted. Every condition, including the QKV baseline, shows $|\Delta\text{Overfit}| \leq 0.0014$ nats (Table 2): validation loss is still flat-to-falling at epoch 100 for all four models. The DeiT-style augmentation stack (RandAugment, Mixup, CutMix, stochastic depth, label smoothing) evidently suppresses overfitting for ViT-S/16 on ImageNet-1k at 100 epochs regardless of attention variant, unlike the 36-layer language-modeling setting in Gao and Xu [2026] where standard attention showed a substantially larger gap (0.8404 vs. 0.6866 for QVV(3)) than any baseline here. Since the baseline itself doesn't overfit, there is no gap for shared-projection attention to reduce, and this experiment cannot distinguish a real absence of the effect from an effect masked by regularization; a lower-augmentation or longer-schedule run would be needed to test H2 properly.

H3 (symmetric vs. asymmetric penalty) — mixed; collapse, not symmetry, looks like the driver. The fully symmetric Q=K=V$^+$ trails the asymmetric variants by 4.91 pts (vs. Q-K=V) and 5.79 pts (vs. QVV(3)) — a substantial penalty for collapsing to a symmetric attention map, running counter to H3's expectation that spatial (non-causal) attention would make symmetry comparatively cheap. The follow-up run (Section 5.2) now supplies the second symmetric variant the original design called for: Q=K-V$^+$, which shares Q and K but keeps V separate, matching Q-K=V's parameter count and cache target exactly. Its gap to Q-K=V is only 1.27 pts — an order of magnitude smaller than Q=K=V$^+$'s 4.91-pt gap to the same reference — despite both Q=K-V$^+$ and Q=K=V$^+$ producing a symmetric attention map. That points at collapsing V into the shared projection, not attention-map symmetry per se, as the larger source of the penalty, which is closer to H3's original motivation (symmetry itself being comparatively cheap for spatial attention) than the verdict the single Q=K=V$^+$ data point supported alone. This reading carries the caveat that Q=K-V$^+$ trained at a different effective batch size than Q-K=V (Section 5.2), so it is suggestive rather than a controlled confirmation, and a same-batch-size rerun is needed before treating H3 as resolved either way.

H4 (detection transfer) — untested. No detection experiment was run this iteration (Section 5); H4 remains open.

7 Limitations

This iteration covers a single architecture (ViT-S/16), a single seed per condition, and a single compressed schedule (100 epochs); none of these are varied, so we cannot report variance across seeds or confirm that findings hold at ViT-B/16 or ViT-L/16 scale, or under the full 300-epoch

DeiT recipe. The ImageNet-100 pilot phase and the ViT-B/16 confirmatory run described in Section 4 were both skipped, as was the downstream detection-transfer experiment (H4). All five originally-planned conditions have now been trained, but not under matched settings: the follow-up run adding Q=K-V$^+$ (Section 5.2) used effective batch 256 (2-GPU DDP, linearly-scaled LR) rather than the batch 128 used for the other four conditions, so its H3 comparison against Q-K=V is suggestive, not a controlled replacement for a same-batch rerun. The same follow-up batch added an unplanned sixth condition, Q=K=V with the positional correction removed, whose large accuracy swing relative to Q=K=V$^+$ is confounded with that same batch-size difference and cannot yet be attributed to the correction itself (Section 5.2). As with Kayyam et al. [2026], the compressed 100-epoch schedule (rather than the full 300-epoch DeiT recipe) means absolute accuracy numbers should not be compared directly to published SOTA ViT results trained with the full recipe or with ImageNet-21k pretraining; the same compression is also the likely cause of the null H2 result above, since it may not train long enough, or with light enough effective regularization net of augmentation, for any condition to visibly overfit.

8 Conclusion

We tested whether projection-sharing attention — validated at small-to-medium scale in Kayyam et al. [2026] and reframed and generalized via depth-m factorization in Gao and Xu [2026] — generalizes to ImageNet-1k, using ViT-S/16 as a first, deliberately narrowed scale point. Q-K=V and QVV(3) recover most but not all of the QKV baseline's top-1 accuracy (2.71 and 1.83 pt gaps respectively), while the fully symmetric Q=K=V$^+$ lags well behind (7.62 pt gap) — a larger, not smaller, symmetric-attention penalty than in the causal language-modeling setting, running against H3's motivating expectation. A follow-up run (Section 5.2) filled in the previously-skipped Q=K-V$^+$ condition and found its gap to the matched-capacity asymmetric Q-K=V is only 1.3 pts, an order of magnitude smaller than Q=K=V$^+$'s — suggesting the earlier verdict conflated symmetry with the separate cost of collapsing V into the shared projection, though this reading is not yet confirmed under matched batch size. The same follow-up unexpectedly found that removing the 2D positional correction from Q=K=V raises its top-1 by about 4 pts, a result we can currently only report, not explain, since it is confounded with that run's larger batch size. The overfitting-reduction effect central to Gao and Xu [2026]'s account of *why* shared-projection attention works did not appear for any condition here, baseline included, because the DeiT augmentation recipe suppressed overfitting broadly at this schedule; this leaves H2 open rather than refuted. The ViT-B/16 confirmatory run, ImageNet-100 pilot, detection-transfer evaluation, and a batch-size-matched rerun of the follow-up conditions all remain as future work.

References

J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebrón, and S. Sanghai. GQA: Training generalized multi-query transformer models from multi-head checkpoints. In *EMNLP*, 2023.

A. Borji. Key-value transformer. *arXiv preprint arXiv:2305.19129*, 2023.

DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024.

A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani,

M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021.

X. Gao and X. Xu. Keyless attention: Value-space routing and value-only caching for efficient transformers. *arXiv preprint arXiv:2606.21848*, 2026.

A. Kayyam, A. Madan Gopal, and M. A. Lewis. Do transformers need three projections? Systematic study of QKV variants. In *Forty-third International Conference on Machine Learning (ICML)*, 2026.

J. Lee et al. Rethinking query, key, and value embedding in vision transformer under tiny model constraints. *arXiv preprint arXiv:2111.10017*, 2021.

Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, 2021.

Anonymous. Share your attention: Transformer weight sharing via matrix-based dictionary learning. *arXiv preprint arXiv:2508.04581*, 2025.

N. Shazeer. Fast transformer decoding: One write-head is all you need. *arXiv preprint arXiv:1911.02150*, 2019.

H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou. Training data-efficient image transformers & distillation through attention. In *ICML*, 2021.

A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In *NeurIPS*, 2017.